

HW 05 SOLUTIONS

Practice Problems

3.37

- a. Not a binomial random variable.
- b. Not a binomial random variable.
- c. Binomial with $n = 100$, $p =$ proportion of high school students who scored above 1026.
- d. Not a binomial random variable (not discrete).
- e. Not binomial, since the sample was not selected among all female HS grads.

3.39

Let $Y = \#$ of components failing in less than 1000 hours. Then, Y is binomial with $n = 4$ and $p = .2$.

- a. $P(Y = 2) = \binom{4}{2}(0.2)^2(0.8)^2 = 0.1536$
- b. The system will operate if 0, 1, or 2 components fail in less than 1000 hours.
 $P(\text{system operates}) = .4096 + .4096 + .1536 = .9728$.

3.51

- a. $P(\text{at least one 6 in four rolls}) = 1 - P(\text{no 6's in four rolls}) = 1 - (5/6)^4 = 0.51775$.
- b. Note that in a single toss of two dice, $P(\text{double 6}) = 1/36$. Then: $P(\text{at least one double 6 in twenty-four rolls}) = 1 - P(\text{no double 6's in twenty-four rolls}) = 1 - (35/36)^{24} = 0.4914$.

3.55

$$\begin{aligned}
E\{Y(Y-1)(Y-2)\} &= \sum_{y=0}^n \frac{y(y-1)(y-2)n!}{y!(n-y)!} p^y (1-p)^{n-y} = \sum_{y=3}^n \frac{n(n-1)(n-2)(n-3)!}{(y-3)!(n-3-(y-3))!} p^y (1-p)^{n-y} \\
&= n(n-1)(n-2)p^3 \sum_{z=0}^{n-3} \binom{n-3}{z} p^z (1-p)^{n-3-z} = n(n-1)(n-2)p^3
\end{aligned}$$

Equating this to $E(Y^3) - 3E(Y^2) + 2E(Y)$, it is found that

$$E(Y^3) = 3n(n-1)p^2 - n(n-1)(n-2)p^3 + np.$$

3.59

If $Y = \#$ of defective motors, then Y is binomial with $n = 10$ and $p = .08$. Then, $E(Y) = .8$. The seller's expected next gain is $\$1000 - \$200E(Y) = \$840$.

Submitted Problems

3.38

Note that Y is binomial with $n = 4$, $p = 1/3 = P(\text{judge chooses formula } B)$.

- $p(y) = \binom{4}{y} \left(\frac{1}{3}\right)^y \left(\frac{2}{3}\right)^{4-y}$, $y = 0, 1, 2, 3, 4$.
- $P(Y \geq 3) = p(3) + p(4) = \frac{8}{81} + \frac{1}{81} = \frac{9}{81} = \frac{1}{9}$.
- $E(Y) = 4 \left(\frac{1}{3}\right) = \frac{4}{3}$.
- $V(Y) = 4 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right) = \frac{8}{9}$.

3.40

Let $Y =$ number that recover from stomach disease. Then Y is binomial with $n = 20$ and $p = .8$.

To find these probabilities, Table 1 in Appendix III will be used.

- $P(Y \geq 10) = 1 - P(Y \leq 9) = 1 - .001 = .999$.
- $P(14 \leq Y \leq 18) = P(Y \leq 18) - P(Y \leq 13) = .931 - .087 = .844$.
- $P(Y \leq 16) = .589$.

3.44

Let Y = number of successful operations, $Y \sim \text{Binomial}(n = 5, p)$

- With $p = 0.8$: $P(Y = 5) = 0.8^5 = 0.328$
- With $p = 0.6$: $P(Y = 4) = 5(0.6^4)(0.4) = 0.259$
- With $p = 0.3$: $P(Y < 2) = P(Y = 1) + P(Y = 0) = 0.528$

3.48

- a. Let Y = number of sets that detect the missile. Then Y has a binomial distribution with $n = 5$ and $p = .9$. Then,

$$P(Y = 4) = 5(.9)^4(.1) = 0.32805$$

and

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - (.1)^5 = 0.99999.$$

- b. With n radar sets, the probability of at least one detection is $1 - (.1)^n$.

If $1 - (.1)^n = .999$, then $n = 3$.

3.50

The only way team A can win in exactly 5 games is to win 3 in the first 4 games and then win the 5th game. Let Y = number of games team A wins in the first 4 games. Thus, Y has a binomial distribution with $n = 4$. Thus, the desired probability is given by

$$P(\text{Team A wins in 5 games}) = P(Y = 3)P(\text{Team A wins game 5})$$

$$= \binom{4}{3} p^3 (1 - p) p = 4p^4 (1 - p).$$

3.56

Using expression for the mean and variance of Y = # of successful explorations, a binomial random variable with $n = 10$ and $p = .1$, $E(Y) = 10(.1) = 1$, and $V(Y) = 10(.1)(.9) = 0.9$.

3.58

If $Y \sim \text{Binomial}(n = 4, p = 0.1)$:

$$E(Y) = 0.4, V(Y) = 0.36, \text{ so } E(Y^2) = 0.36 + (0.4)^2 = 0.52$$

$$\text{Then } E(C) = 3(0.52) + (0.36) + 2 = 3.96$$